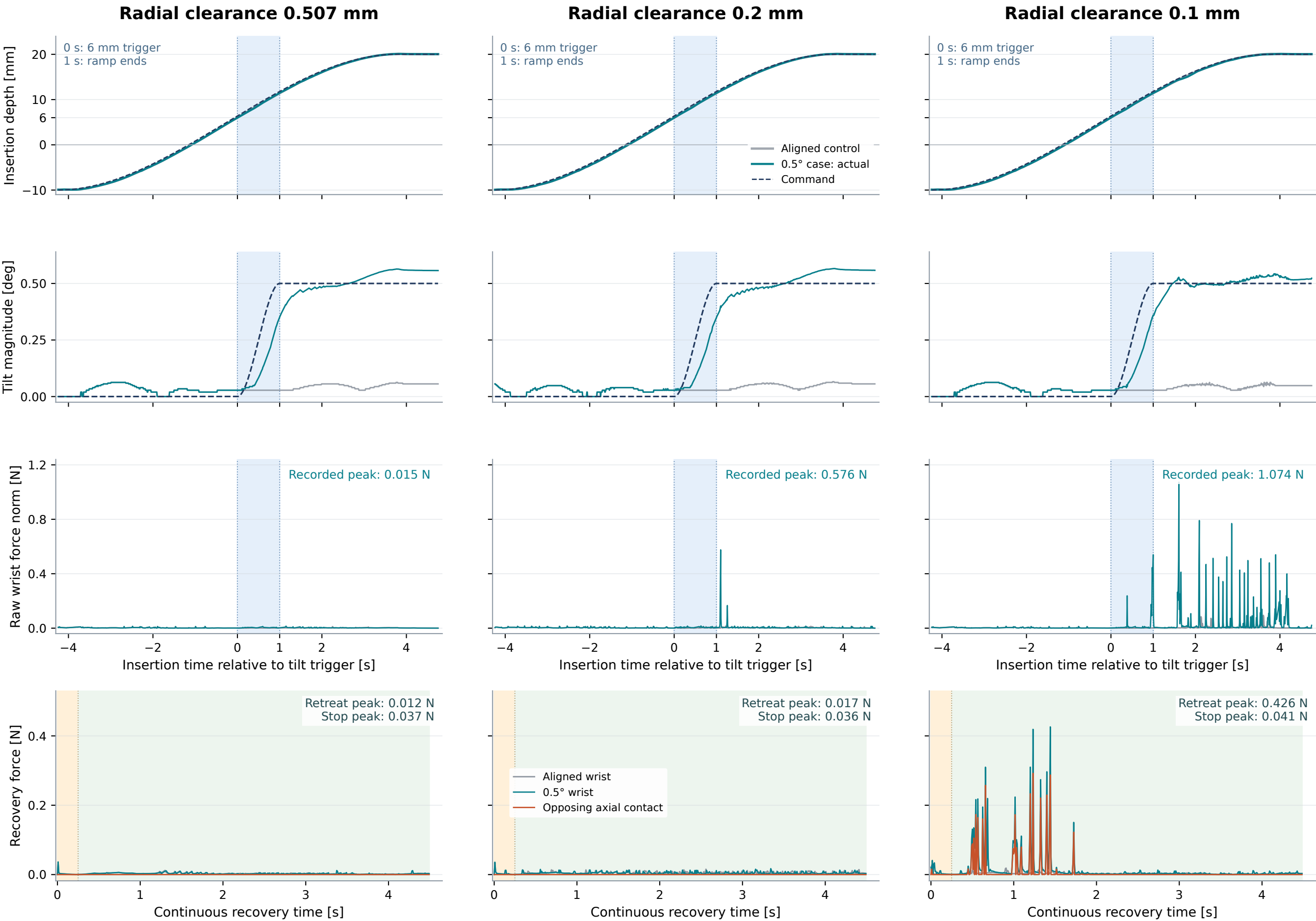


Smaller clearance raises loads under a mild insertion tilt

0.5° pitch starts at 6 mm actual depth (30% of a 20 mm insertion); smooth 1 s ramp.

All three tilted insertions and continuous straight withdrawals succeeded; no insertion stall was detected.



Blue shading: tilt ramp. Amber: 0.25 s stop. Green: direct withdrawal until clearance. The recovery plots use the original continuous motion, not replay probes.

Wrist values are raw reaction-force norms; opposing axial contact = $\max(0, -\text{world contact } F_z)$. Common force scales are used across gaps. Numerical screening passed; convergence is not established.